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Social identification effects in product and process development teams

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Abstract

This study integrated theories of group effectiveness with social identity and social categorization theory to develop and test a model of team performance. The sample included 42 product and process development teams employed in three divisions of a Fortune 500 manufacturing firm. Performance data was collected at project midpoint and at project completion through both managers' and team members' assessments. The study found that team social identification was an important predictor of team performance, explaining variance beyond that explained by cohesiveness and external communication. Team social identification was influenced by top management support and recognition, project leader organizational influence, the extent to which members' time was dedicated to the team, and team functional diversity. Implications for scholars and professional managers are discussed. © 1997 Elsevier Science B.V.

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1. Introduction

Today, task forces and project teams are recognized as important mechanisms for the effective management of nonroutine problems in organizations (Gersick, 1988). Project teams allow organizations to reconfigure their internal structures on an 'as needed' basis, mobilizing unique combinations of differentiated knowledge and expertise, and simultaneously providing closer coupling of functional areas (Ancona, 1990). They provide flexibility, increased responsiveness to external conditions, and tighter control of resources through scheduling at the project level (Katz and Allen, 1985).

Arguably, temporary teams are of the greatest benefit to organizations in the context

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of product and process development. Many researchers have suggested that new products and processes are critical to the long-term survival of organizations (e.g., Clark and Fujimoto, 1990, 1991; Ancona and Caldwell, 1987), and teams have recently been called 'the heart of efficient product development' (Brown and Eisenhardt, 1995, p. 369). Development teams are task forces assembled to bring a new product or process from conception through design, prototype or pilot, and into production. Because of their promise and prevalence, development teams have received substantial research attention to date. Studies of product development projects have found team effectiveness related to the frequency and content of external communications (Ancona and Caldwell, 1992a; Katz and Tushman, 1979; Allen, 1971, 1977), internal cohesiveness and communications (Dougherty, 1990, 1992; Keller, 1986), team tenure (Katz, 1982), powerful project leaders (Clark and Fujimoto, 1991; Katz and Allen, 1985), and top management political and resource support (e.g., Cooper and Kleinschmidt, 1987; Katz and Allen, 1985). While these findings have enhanced our understanding of both context and process influences on team performance, numerous aspects of team functioning remain to be explored.

The current study is concerned with the issue of team identity and explores the significance of members' social identification to the effectiveness of project development teams. The concept of group identity is an old one and underpins much of the theoretical literature on group effectiveness. Team development practitioners, as well as most group theorists have long posited that having a strong group identity is a necessary condition of group effectiveness (e.g., Alderfer, 1987; Hackman, 1987; McGrath, 1984). They argue that when group boundaries lack clarity and become overly permeable, the group looses its identity and, subsequently, its ability to integrate and work as a group. However, up until quite recently, issues of identity have been of little concern in the laboratory studies that have dominated much of small group research (Sundstrom and Altman, 1989).

Further, even in more recent field studies, the degree to which members socially identify with teams has seldom been empirically examined (Campion et al., 1996). This is perhaps due to the fact that most group effectiveness research has been conducted with stable, long-standing work teams engaged in repetitive tasks (cf. Hackman, 1987). Such teams have shared histories with well-developed cultures, members whose time is allocated 100% to the group, and readily understood and publicly acknowledged purposes and status within the organization. As importantly, ongoing work groups are embedded in static bureaucratic structures with fixed boundaries that clearly distinguish them from other groups in the organizational setting. In the current paper, I argue that identity issues are particularly problematic for project teams due to the diversity of their membership and the determinacy of their existence (Emmanuelides, 1993).

Development teams are typically cross-functional: members are representatives of various organizational subunits who have been 'loaned' to a project due to specific expertise. Functional diversity (a) brings a variety of perspectives and differing knowledge to bear on design issues, (b) helps to forestall downstream manufacturing and marketing problems (e.g., Clark and Fujimoto, 1991; Nonaka, 1990), and (c) encourages external communication (Ancona and Caldwell, 1992b) which, research suggests, is critical for team success (Ancona and Caldwell, 1992a; Katz and Tushman, 1979, 1981;

Allen, 1971, 1977). Yet, functional diversity has been found to relate negatively to the technical quality of development projects (Ancona and Caldwell, 1992b), perhaps because the different 'thought worlds' of members from different functions make the integration of members' ideas and perspectives difficult (Dougherty, 1992). Further, the primary social identity and allegiance of members from different functions usually resides with their home subunit rather than with the project team (Alderfer, 1987), particularly when members are aware that they will return to the home unit at the end of the project. Recent case studies have demonstrated that team membership, identity, and loyalty are problematic in project teams (Donnellon, 1993; Gersick and Davis-Sacks, 1990).

Identification has most often been conceptualized in organizational studies as value congruence between members and their group, or internalization of a group's values by members (e.g., Chatman, 1991; O'Reilly et al., 1991; Hall and Schneider, 1972). In contrast, the social identity (Tajfel and Turner, 1985; Turner, 1982) and social categorization perspectives (Turner, 1982, 1987) define identification as the cognitive link between individuals and groups arising from the images or mental representations that members have of the group. A growing theoretical literature suggests that the members' cognitive identification with a group has a profound influence on their behavior and, subsequently, on group outcomes (Dutton et al., 1994; Ashforth and Mael, 1989). For example, when members are socially identified with a group, they tend to adopt the group's central problems as their own and cooperate in their resolution, regardless of their feelings toward other group members (Brewer and Gardner, 1996). Given its potency to explain group behavior, it is important to investigate antecedents of social identification in project teams.

2. A theoretical model of team performance

In this study, I proposed a model of team performance that integrates theory and research from social identity and social categorization theory with the existing research literature on internal and external group processes. The theoretical model (Fig. 1) is presented here to make it easier to follow the discussion pertaining to each of the hypotheses contained within it. The model specifies that team members' social identification, team cohesiveness, and the amount of external communication jointly influence project team performance. Team social identification is posited to result from the joint influences of functional diversity, members' time dedication to the project, project leaders' organizational influence, and top managers' support and recognition of the project. The theory and hypotheses relative to cohesiveness and external communication have been tested previously in the context of development teams. By testing a model that predicts their joint effects with team members' social identification, I investigated the extent to which social identification adds to our current understanding and explanation of team performance.

Prior to stating the hypotheses, it is important to note that team performance itself is multidimensional and is most often in the eye of the beholder (Tsui, 1984). For example, performance criteria in studies of production teams have included measures of produc-

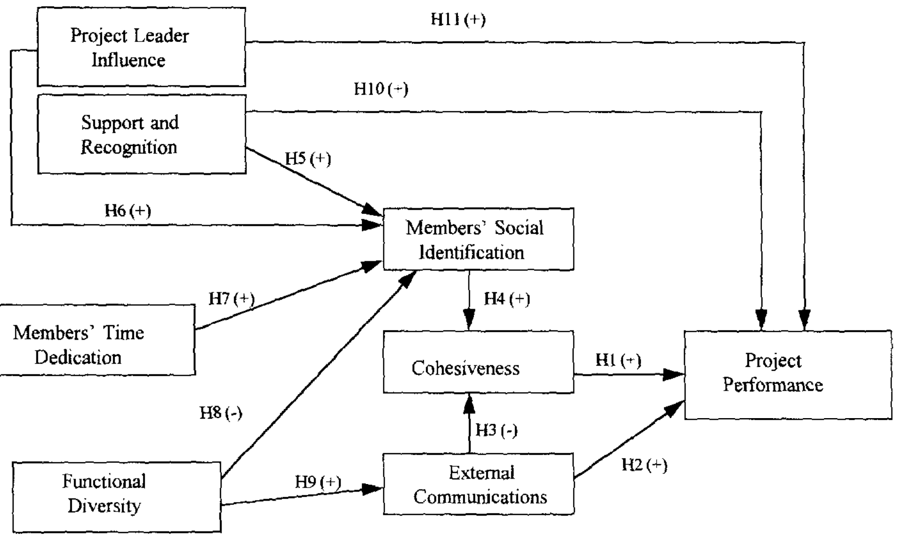

Fig. 1. Hypothesized model of project performance.

tivity, absenteeism, or turnover. In other types of teams, different measures might be more appropriate both to the research setting under study and to the nature of the research question. Hackman (1987) argued that performance criteria in the study of groups should include three criteria: (1) client satisfaction; (2) goal attainment; and (3) team member satisfaction, and most research studies of group performance now use multiple performance criteria. For example, in the R&D setting, recent studies of team performance have used criteria developed through collaboration with R&D managers including adherence to budgets and schedules, and project technical quality (Ancona and Caldwell, 1992a,b; Keller, 1986). Further, prior empirical work has demonstrated that team member assessments of performance are seldom congruent with manager ratings (Ancona and Caldwell, 1992a; Keller, 1986) or with objective measures (Gladstein, 1984). Consistent with prior studies in R&D, this study used both managers' and team members' assessments of team performance.

2.1. Team performance and team process

Early theory and research on group functioning was largely the province of social psychologists who tended to focus on internal group processes (Gladstein, 1984). This research tradition provided fine-grained analyses of dynamics occurring within a group's boundaries and, particularly, of interpersonal relations within the group. Others have noted that integration of members within a group is not so much a function of role requirements as it is of affective factors, such as interpersonal attraction and satisfaction with other group members (Katz and Kahn, 1978). Turner (1982) labeled research in this tradition 'the Social Cohesion model' as its underlying premise is that team members must form a cohesive unit to perform adequately. Cohesiveness reflects interpersonal interactions and attraction among team members, is primarily affective in nature, and develops vis a vis the relationships among team members (e.g., O'Reilly et al., 1991; Shaw, 1971; Lott and Lott, 1965). Empirical support for the positive effects of cohesiveness on R&D team performance has been reported in several studies (Keller, 1986; Pelz and Andrews, 1976; O'Keefe et al., 1975). Thus, I proposed the following hypothesis:

Hypothesis 1. *There is a positive relationship between project team performance and cohesiveness*

With increasing emphasis on open system models (Katz and Kahn, 1978), group theorists and researchers began to turn their attention to the context in which groups operate. For example, research from the socio–technical systems school focused on the interrelationships of group process and context (Cummings, 1978). Work on boundary spanning investigated how groups regulate the flow of information across their borders (e.g., Tushman, 1979). Most boundary spanning studies have been conducted in R&D settings with product development groups. The need for information exchange between development groups and other groups within the organization has been identified by numerous authors (Ancona and Caldwell, 1987; Burgelman, 1983; Quinn and Mueller, 1963). Development teams must interact with top management to garner continued support for the project, coordinate with other subunits in the organization, and scan the environment for relevant technical information (Ancona and Caldwell, 1992a). It is also not unusual for such teams to have significant communication with customers and suppliers outside of the organization. Numerous studies in R&D settings report that the amount of information exchange or communication among a team and other groups external to the team is positively related to development team performance (e.g., Ebadi and Utterbach, 1984). In fact, a number of recent studies found that the amount of external communication was more important to manager-rated team performance than were internal task processes (Ancona and Caldwell, 1992b; Gladstein, 1984). Thus, from prior theory and research, I posited that:

Hypothesis 2. *Team performance is positively related to the amount of external communication*

The relationship between group internal processes and external communication has also received some discussion in the literature on groups. Janis (1982) noted that, under conditions of extreme cohesiveness, groups establish gatekeeping roles to close off their boundary to outside communication. Others have noted that when groups import information on differing goals, perspectives, and attitudes, intragroup conflict is likely to ensue (Shaw, 1971; Schmidt and Kochran, 1972). Recently, Ancona and Caldwell (1992a) found that cohesiveness was negatively correlated with frequency of external communication in a study of development groups. Thus, I proposed:

Hypothesis 3. *Cohesiveness is negatively related to the amount of external communication*

2.2. *Team identity and members' social identification*

Numerous researchers have argued that systems with boundaries that are too permeable lose their sense of identity and their ability to perform as a group (e.g., Alderfer, 1987; Friedlander, 1987) Many boundaries in the workplace are physical (e.g., white lines painted on a shop floor to indicate the private domain or territory of the assembly department). But, in a broader sense, boundaries may be anything that makes salient to members and outsiders characteristics differentiating a group from other groups within the social environment (Sundstrom and Altman, 1989). Differentiation is critical to identity formation whether at the individual or group level.

Tajfel (1978), in studies of intergroup conflict, argued that group identities were constructed of perceived prototypical characteristics of members of a group. Alternatively, other researchers note that group identity also includes aspects of a group that are independent of individual members such as group goals, mission, practices, values, and actions (Ashforth and Mael, 1989; Albert and Whetten, 1985). For example, organizational identity has been defined as a cognitive representation held by organizational members of that which is central, distinctive, and enduring about an organization (Dutton et al., 1994; Albert and Whetten, 1985). Similarly, a team identity is formed from central and distinctive aspects of the project team, which may include the project mission or purpose, goals, actions and prototypical aspects of team members themselves. Team identity is a valuable concept, inasmuch as it helps explain social identification of individual group members. While perceptions of identity vary across team members, as they interpret what is distinctive about a group based on their individual needs and unique prior experiences, social identification processes operate independently of identity content (Tajfel, 1982).

Social identity theory has proven particularly useful for understanding member identification in large social groups and identity groups (e.g., racial groups, gender groups) where interaction is generally constrained to a small subset of members or does not occur at all (Tajfel, 1978, 1982; Turner, 1985, 1987). To date, there has been research in organizational settings focused on peoples' social identification with their employing organization (Dutton et al., 1994), their alma mater (Mael, 1988), and their occupational group (Van Maanen and Barley, 1984). Perhaps because of the relatively less enduring, more ephemeral nature of project teams and other ad hoc groups, there has been little empirical attention thus far given to performance effects of social identification among members. Thus, the usefulness of social identity theory for explaining performance in smaller face-to-face work teams has not yet been fully explored.

Social identity is best understood as one aspect of the self-concept that, along with the more idiosyncratic personal identity, provides people with an answer to the question 'Who am I?' Where the personal aspect of identity might be thought of as a composite image of past experiences and personal history, a person's social identity is constructed with reference to membership in different social categories. These social categories can be ascribed (e.g., females, Americans, etc.), self-selected (e.g., professors, accountants), or the result of organizational categorization processes (e.g., member of the New Generation Project team). Social identity suggests to an individual where he or she 'fits' in the social milieu. Social identification is the internalization of a social identity and involves the perception that one is '... psychologically intertwined with the fate of the group ... personally experiencing the successes and failures of the group' (p. 21), and defining oneself in terms of the group (Ashforth and Mael, 1989). In short, identification represents the psychological acceptance of group membership.

It is important to note that, from the social identity perspective, identification does not originate from interpersonal relationships among group members (Turner, 1985), a basic premise of the social cohesion model presented earlier. People are often socially identified with groups whose specific members have little contact with them (Turner, 1982). The social connections inherent in social identity theory are not personalized bonds of attachment to others, but '... impersonal bonds derived from common identification with some symbolic group or social category' (Brewer and Gardner, 1996, p. 83). Thus, social identity theory draws our attention away from interpersonal relationships among team members to cognitive representations of a group based on other criteria (Hogg, 1992).

Social identification is also distinct from commitment to a group. While organizational commitment has been defined as the strength of an individual's identification with an organization, it is composed of a belief in and acceptance of an organization's values, willingness to assert effort on behalf of an organization, and a desire to maintain membership (Mowday et al., 1979). Mael and Ashforth (1995) argue that 'this view includes internalization of values, behavioral intentions, and positive affect, but not identification' ... the cognitive perception of oneness with an organization' (p. 312). Mael and Tetrick (1992), using confirmatory factor analysis, found that social identification was empirically distinct from organizational commitment, and that social identification had lesser conceptual overlap with other measures of effect (e.g., job satisfaction).

However, as a social identity becomes more salient, attention to individual goals and interests gives way to group goals and interests (Kramer and Brewer, 1984), indicating that social identification positively influences commitment. Further, interaction processes in socially identified groups improve as members adopt a cooperative orientation toward resolution of group problems. This is not the result of liking among the group members. Rather, it reflects the members' heightened interest in the welfare of the symbolic group with which they are identified. As Turner (1982) noted, people do not so much identify with those they like, as they begin to like those with whom they identify. Research on social identity theory has shown that social identification is positively related to preferential treatment (Tajfel and Turner, 1985), greater cooperation and interaction (Dutton et al., 1994; Ashforth and Mael, 1989), more positive attitudes (Turner, 1982), and more helpful behaviors (Dutton et al., 1994; Ashforth and Mael, 1989; Turner, 1982) among in-group members relative to out-group members. Further, once socially identified with a group, individuals accentuate the similarities among 'in-group' members while simultaneously accentuating the differences between the 'in-group' and the 'out-group' (Hogg and Abrams, 1985). Thus, as identification increases among team members, we expect to see more intrateam cooperation contributing to increased cohesiveness among members.

Hypothesis 4. *Cohesiveness is positively related to team members' social identification.*

2.3. Team structure and context

The previous sections discussed the expected interrelationships among social identification, cohesiveness, and external communication and their joint effects on team performance. This section elaborates on several structural and contextual aspects of team functioning that are posited to influence each of the process variables and, subsequently, team performance.

Within organizations, people have multiple organizationally defined identities. They are members of functional areas (e.g., the production function, the R&D function), of professions or occupational groups, (e.g., lawyers, scientists, technicians), of intact work groups (e.g., the applied research unit, the chemistry laboratory subunit), as well as of temporary project groups. Each of these identities may be more or less salient depending on situational as well as temporal cues (Kramer, 1991). Several theorists (Ashforth and Mael, 1989; Kramer, 1991) have argued that identification at the level of the primary work group (e.g., department or subunit) is the most salient because member similarity, task interdependence, and interpersonal proximity are all greater with respect to the primary work group. Each of these aspects of primary groups, along with their enduring nature, tend to solidify their identity within an organization. As importantly and often overlooked, primary group identities are made public knowledge and reinforced through formal communication processes, the use of which are ubiquitous in most organizations (e.g., organizational charts). However, in today's fluid, team-based organizations, the primacy of the functional department in identity formation is questionable as people spend greater amounts of time in various task groups. To the extent that task teams develop distinguishable identities, social identification processes are expected to occur among team members.

As previously mentioned, particular characteristics of temporary project teams make establishing an identity challenging. Members are typically chosen for the express purpose of bringing multiple divergent perspectives to bear on unstructured problems of organizational significance. Thus, they are likely to have strong ties to their functional departments. In addition, members are typically on loan to such teams with expectations of returning to their 'home' group on completion of the task. The psychological start-up costs involved in accepting a new social identity may be perceived of by team members as simply too high given the limited life expectancy of the team. Establishing a team identity might be viewed, therefore, as one of the largest challenges a new team faces, and it is ironic that so little attention is paid to this process in the literature (Kramer, 1991).

There is substantial evidence from laboratory studies that the mere act of categorization induces people to internalize a group membership no matter what the effect on their self-perception (Tajfel and Turner, 1985). Individuals may try to rationalize their assignment to specific groups by assuming that they possess prototypical characteristics of the group or category to which they have self-selected or been assigned (Kramer, 1991). This has been called self-stereotyping (Turner, 1982) and self-categorization (Ashforth and Mael, 1989). However, Schlenker (1985) and Turner (1982) argued that social identification is not deterministic, as individuals, to some degree, choose and negotiate social identities (Swann, 1987) and cognitively shift among identities to meet their individual needs at any given time (Hogg and Abrams, 1985).

The social identity literature suggests several factors that are likely to increase an individual's identification with a group. The first is the distinctiveness of a group's values and practices (Ashforth and Mael, 1989; Mael and Ashforth, 1992; Tajfel and

Turner, 1985). Individuals seek to enhance their sense of self-importance by affiliating and identifying with groups that provide them with social identities that accentuate their own distinctiveness (Dutton et al., 1994). To an even greater extent, individuals identify more with social identities that enhance their self-esteem than to others that might be available (Dutton et al., 1994; Ashforth and Mael, 1989; Mael, 1988).

Individual self-esteem results from self-evaluation processes against a set of internal criteria or standards. However, the criteria against which people judge their own adequacy are themselves internalized from the embedding culture (Erez and Earley, 1993; Bandura, 1986), suggesting that self-evaluations largely reflect social information. Thus, social identification is likely to increase with perceptions of the valence an embedding culture places on a specific social identity. To the extent that valence is represented by a group's prestige, social identification is posited to vary with the prestige of the group (Ashforth and Mael, 1989), and empirical support has been reported for this relationship (Mael, 1988). In the context studied here, team members should identify more with prestigious projects that have publicly acknowledged value within the organization. Team members infer the value placed on a particular project from several cues, three of which were investigated here. These included (1) top management support to and recognition of a project; (2) the influence or power of the project leader; and (3) the extent to which team members' time is dedicated to the project.

At any given time within an organization, multiple development projects exist simultaneously, each with a claim on limited organizational resources. The allocation of resources and recognition is the province of top management, and it serves symbolic as well as instrumental functions. Both resource-based support and public recognition of a team's efforts convey the importance attributed by top managers to a project, and, subsequently, through identification, the importance and significance members ascribe to the team. Further, to the degree that the level of top management support and recognition are recognized by other organizational participants, the prestige of a project and the project team increase throughout the organization.

Prestige is also likely to be attributed to teams that have influential project leaders. Project leaders have the responsibility for integrating the work of project members from different functions and technical specialties. Katz and Allen (1985) noted that leaders of highly effective projects have strong organizational influence outside of the project, and numerous studies support this relationship (e.g., Katz and Allen, 1985; Clark and Fujimoto, 1991). Katz and Tushman (1981) found that team members' motivation and behaviors are affected both by leadership within the team and by organizational influence outside the team. The assignment of organizationally influential project leaders (self-selected or chosen by management) causes team members and outsiders to infer that the project is important and prestigious and, subsequently, enhances the self-esteem of team members.

Further, it is not unusual in product development settings for team members to be simultaneously serving on multiple projects or simultaneously engaged in functional duties (Donnellon, 1993; Dougherty, 1992). Multiple memberships can serve to blur the boundaries of a team, thereby reducing its distinctiveness (Ashforth and Mael, 1989). As importantly, when members' time is dedicated to a project, it signals the relative value or prestige of the project to both team members and other organizational participants. In addition, the more team members' time is allocated to a project, the greater is the opportunity for team members to interact with each other. As previously mentioned, individuals tend to accentuate differences between groups with which they identify and out-groups (Hogg and Turner, 1988), dividing the world into 'we' and 'them' and assuming that others are representative of their group. Increased interaction tends to weaken the perception of other members as representatives of their own functional group and strengthen the perception of them as one of 'us'. To the extent that members' time is dedicated to a project, members will become more aware of their shared common fate and less aware of their differences.

To summarize, to the extent that contested organizational resources—such as top management support and recognition, powerful project leaders, and member time—are allocated to a project, individuals tend to attribute greater prestige to the project team. Since individuals identify more with prestigious groups in the interest of enhancing their own self-esteem, I proposed:

Hypothesis 5. *Team members' social identification is positively related to top management support and recognition.*

Hypothesis 6. *Team members' social identification is positively related to the organizational influence of the project leader.*

Hypothesis 7. *Team members' social identification is positively related to the degree to which their time is dedicated to a project.*

Finally, cross-functional development teams vary in their degree of functional heterogeneity. Some teams may be composed predominantly of members from one subunit such as R&D, with only one or two members from manufacturing or marketing. Other teams may be composed of members, each of whom is from a different subunit. As discussed earlier, team members on loan from functional subunits may be strongly identified with the 'home' unit to which they will return on the completion of the project. Functional unit identity is strengthened by member similarity in education, training, and goal orientation, task interdependence, and interpersonal proximity (Byrne, 1971; Kramer, 1991). Thus, to the extent functional diversity exists in a team, team social identification is likely to be reduced.

Hypothesis 8. *Team members' social identification is negatively related to functional diversity.*

Functional diversity is also posited to have an indirect effect on team performance through external communication. At the very least, team members from disparate functions will have previously established relationships with individuals from their home unit with whom they continue communicating. This is, in fact, the means through which the promised closer coupling of functional areas occurs. In addition, similarity encourages communication and interaction (Byrne, 1971), and to the extent that individuals are similar in goals, orientations, education, and training to people in their home unit, we expect to see increasing communication with the home unit. Zenger and Lawrence (1989) found that similarity of tenure was related to the amount of communication between team members and non-members. Others have also reported a positive relationship between functional diversity and external communications (Ancona and Caldwell, 1992b).

Hypothesis 9. *The amount of external communication is positively related to functional diversity.*

A number of the structural and context variables contained in the model are also expected to have direct effects on team performance independent of the members' social identification. Specifically, top management support provides additional resources for a project that have been shown by others to be related to group outcomes (e.g., Gladstein, 1984), and recognition is likely to predispose other organizational units to cooperate with a team. Similarly, organizationally influential project leaders help secure 'essential resources, mediate intergroup conflicts, and ... protect the developmental effort from outside sources of interference' (Katz and Allen, 1985, p. 72). Thus, the benefits to a project team of support and recognition and a powerful project leader are instrumental, as well as symbolic. In contrast, although members' time is an equally valued resource, I expected that social identification is a necessary condition for members to exert effort toward the group goal. Thus, I expected no direct influence from dedication of members' time to performance.

Hypothesis 10. *Team performance is positively related to top management support and recognition.*

Hypothesis 11. *Team performance is positively related to the organizational influence of the project leader.*

3. Additional influences on project team performance

There are numerous other characteristics of the team and the task that have been shown to influence project performance in the literature. While it was not possible to include all these variables in the current study, I did control for three variables that prior research suggests may be important to group effectiveness. These included team tenure, newness of the product or process technology to the organization, and team size. Team tenure has been reported to have a curvilinear relationship with project performance in prior studies in R&D settings (Katz, 1982; Smith, 1970). Group size has been shown in numerous studies to influence performance through its effects on communications and group process (Katz, 1982). Finally, increasing uncertainty related to technology creates different requirements for external communication in project teams, and subsequently has implications for team performance (Brown and Utterback, 1985).

4. Research design and measures

4.1. Procedure and sample

I studied product and process development projects from three large manufacturing divisions of one Fortune 500 company. Assessments of the study variables were obtained via questionnaires administered to project team members. At the time the questionnaires were first administered, all of the projects chosen for study were at about the midpoint of their originally scheduled time frames. Fifty-two project teams were originally chosen for study from company records. Thirty of the teams were engaged in product development, and the remaining 22 teams were engaged in process development activities. For all project groups, the focus of their work activity was the use of existing knowledge (although not necessarily at the firm) to create new products and processes.

Each of the 52 project team leaders was sent a letter from the president of their respective divisions explaining the nature of the study. I contacted each project leader by telephone to verify the membership of his or her team. Questionnaires were sent to each team member with a letter encouraging them to voluntarily complete the questionnaire on normally scheduled work time and asking that they return it via mail directly to the study investigator in an enclosed self-addressed envelope. Participants were also assured complete confidentiality related to their individual responses. Each questionnaire was identified with a specific project number and name, and participants were asked to complete the questions in relation to that project. A total of 296 questionnaires were sent out, and 263 (88.9%) were returned.

Since the unit of interest was the project, projects were only included if at least 75% of team members assigned to it responded to the questionnaire. This reduced the number of teams to 48 but missing data in the questionnaires further reduced the total teams used in the study to 42 with a combined membership of 236 individuals. The teams ranged in size from 3 to 7 members with an average size of about 5 members. Team tenure varied from 6 months to 13 months. The individuals on the teams had an average organizational tenure of 14.5 years, an average age of 42.3 years, and were 96% male. In addition, about 83% had at least a four-year college degree, and approximately 25% had graduate degrees.

At the same time a questionnaire was sent to project team members, a separate questionnaire to assess team performance data was sent to functional managers, a division vice-president, and division directors. Further, managerial ratings of team performance were also collected via questionnaire one year after the initial data collection. At the time of the second data collection, 36 of the projects were officially completed, and the remaining six were in the final months of completion.

4.2. Measures

Whenever possible, previously existing measurement instruments with adequate psychometric properties were chosen for use in this study. However, several scales were developed for use here. Since the unit of analysis in this study was the group, items were specifically constructed to refer to the team or to the team's work. Unless otherwise noted, all measures used 5-point Likert response scales ranging from (1) highly disagree to (5) highly agree.

The individual responses were aggregated to the project group level by computing the mean across project members. To justify this aggregation, one-way analyses of variance were conducted to determine if there was greater variability between projects than within projects (Winer, 1971). In all cases, the variance between projects was greater than the variance within, suggesting that aggregation was appropriate. We also calculated an interrater reliability coefficient using the method recommended by James et al.

(1984a) to examine the intragroup reliability of responses. In all cases, the interrater reliability score revealed homogeneity among the team members in their responses to the items in the scales. Results of the F-tests from the analysis of variance and the interrater reliability scores are reported below. In addition, homogeneity of variance was tested using Bartlett-Box F. All of the variables passed this test.

4.2.1. Members' time dedication

Each team member and the project leader was asked to indicate on the questionnaire the percentage of their total work time that was allocated to a focal project. In addition, as a validity check on this assessment, each project leader was asked to indicate the percentage of each member's total work time that was spent on a project. Interrater reliability between the two assessments was 0.76. Since it seemed more reasonable for team members to have greater knowledge of their time allocation across projects, I used the team members' assessments for this variable. The higher the score on this measure, the greater the extent to which members of the team were dedicated to the focal project.

4.2.2. Top management support and recognition

A 4-item scale was constructed for use in this study. Team members and the project leader were asked to indicate the extent to which they believed top management: (1) recognized the importance of the project to the organization; (2) valued the contribution of the team to the project; (3) allocated adequate resources to accomplish the project; and (4) publicly 'talked up' the project to others in the organization. Higher scores indicated greater top management support and recognition. Cronbach's α in this sample was 0.78. One-way analysis of variance found aggregation to the group level appropriate ($F = 3.79$; $p < 0.001$). Finally, the average intragroup reliability for this scale was 0.92.

4.2.3. Leader organizational influence

A 3-item scale was constructed for use in this study. Team members were asked to indicate the degree to which they thought their project leader (1) was generally respected in the organization for his/her expertise and competence, (2) had substantial influence with functional managers in the organization, and (3) had substantial influence with top management in the organization. The project leader was also asked to rate himself or herself on each of the items, and the average rating across the team members and project leader were used. Higher scores indicated greater organizational influence. Cronbach's α in this sample was 0.88. One-way analysis of variance found aggregation to the group level appropriate ($F = 4.02$; $p < 0.001$). Finally, the average intragroup reliability for this scale was 0.94.

4.2.4. Functional diversity

This variable was assessed using heterogeneity index of Blau, 1977: $(1 - \Sigma i^2)$, where i is the proportion of the group in the ith category. Team members' and the project leader's functional affiliations were acquired from company records and verified with project leaders. A high score on this index indicates variability in the functional affiliations of team members or greater functional heterogeneity.

4.2.5. *Members' social identification*

A 7-item scale was used to assess this variable. Four of the items were adapted from Mael and Tetrick (1992) organizational identification scale and 3 additional items were written specifically for this study. The questionnaire indicated, "In this section, we are specifically interested in team members' feelings about being on the project team". Team members and the project leader were asked to indicate the extent to which: (1) team members view criticism of the XXX project team as a personal insult; (2) team members are interested in what others think about the XXX project team; (3) team members consider the success of the XXX project team their success; (4) team members take praise of the XXX project team as a personal compliment; (5) working on the XXX project team defines 'who team members are' at the company; (6) for team members in general, to think of themselves as employees of the company is to think of themselves as members of the XXX project team; and (7) for team members in general, their self-worth at work is related to the performance of the XXX project team. Cronbach's α in this sample was 0.92. Higher scores indicated greater identification with the project.

4.2.6. *Cohesiveness*

A 4-item scale developed by Seashore (1954) was used in this study. The questionnaire indicated, "In this section, we are interested in the relationships among the project team members". Then, team members and the project leader were asked to rate the extent to which team members: (1) are likely to defend each other from criticism by outsiders; (2) help each other while working on the project; (3) get along well with each other; and (4) stick together. The response format was a 4-point scale ranging from 'not very good' to 'great, couldn't be better': higher scores indicated greater group cohesiveness. Cronbach's α in this sample was 0.87. This α is consistent with previously reported research (e.g., Keller, 1986; O'Reilly et al., 1989). One-way analysis of variance found aggregation to the group level appropriate ($F = 3.52$; $p < 0.01$). Finally, the average intragroup reliability for this scale was 0.87.

An exploratory factor analysis, shown in Appendix A, was used to establish the discriminant validity among the conceptual domains represented by cohesiveness and social identification. To demonstrate independence, the individual-level scores for each of these variables were submitted to a principal-components factor analysis. With a two-factor constraint, the rotated factor matrix showed all items loading on the appropriate factor and minimal overlap of items between factors. I concluded from this analysis that team members conceptually distinguished among these constructs.

4.2.7. *External communication*

This variable was assessed using a single item measure consistent with prior research (Ancona and Caldwell, 1992b). Team members and the project leader were asked to indicate how often on average they communicated regarding the project with nonteam members in marketing, manufacturing, engineering, product or division management, outside vendors, or outside customers. The mean across these 6 categories was taken for each team member, and then team members' scores were averaged to develop a team score.

4.2.8. Team performance

This variable was measured using a 5-item scale developed by Ancona and Caldwell (1992a). Team members and the project leader assessed performance at Time 1 (the midpoint of the project) by rating efficiency, quality of technical innovations, adherence to schedules, adherence to budgets, and ability to resolve conflicts. A principal components factor analysis with varimax rotation was conducted on the data, and all items loaded on one factor. This factor was named member-rated performance. Cronbach's α in this study was 0.89. One-way analysis of variance found aggregation to the group level appropriate ($F = 4.89$; $p < 0.001$). Finally, the average intragroup reliability for this scale was 0.85.

Assessments of project performance were also collected from divisional vice-presidents, directors, and functional managers at the same time as team members' ratings. In all cases, at least two and as many as five managers assessed performance on the same criteria as team members. Principal components factor analysis with varimax rotation was conducted on the data, and both the Kaiser criteria and the scree plot suggested that two factors be maintained. One factor was defined by the items relating to adherence to budgets and adherence to schedules and was labeled budgets and schedules ($\alpha = 0.77$). The second factor was defined by the quality of technical innovations and efficiency and was labeled technical quality ($\alpha = 0.83$). The ability to resolve conflicts loaded by itself on a third factor that had an eigenvalue less than 1. As a consequence, it was dropped from subsequent analysis. The results of this analysis were compatible with those reported by Ancona and Caldwell (1992a) except for the conflict item, which loaded on the same factor as quality and efficiency in their study. The average interrater reliability among managers for budgets and schedules was 0.82 and for technical quality was 0.79. One-way analysis of variance found aggregation to the group level appropriate for budgets and schedules ($F = 4.73$; $p < 0.001$) and for technical quality ($F = 3.83$; $p < 0.001$).

Finally, managers' ratings of budgets and schedules and technical quality were collected a second time approximately one year after the initial questionnaire administration. Cronbach's α for budgets and schedules and technical quality from the time 2 ratings were 0.87 and 0.83, respectively. The average interrater reliability among managers for budgets and schedules was 0.89 and for technical quality was 0.91. One-way analysis of variance found aggregation to the group level appropriate for budgets and schedules ($F = 4.65$; $p < 0.001$) and for technical quality ($F = 4.23$; $p < 0.001$).

4.2.9. Team tenure

This was the length of time that the project had been underway, and was acquired from the company records. The measure was then verified with project leaders.

4.2.10. Team size

This was acquired from project documents and verified with the project leader.

4.2.11. Technology newness

This information was acquired from company records. Prior to the inception of a project, the intended project leader was responsible for assessing the extent to which the project involved technology that was new to the organization on a 3-point scale. This assessment was used during the formal justification process for the project, and the directors indicated that the assessment actually represented a consensus arrived at between the project leader and themselves. A rating of 1 indicated that the organization had little or no familiarity with the technology, and a rating of 3 indicated that the organization was very familiar with the technology.

4.3. Analyses

Descriptive statistics and intercorrelations among the variables were computed. Multivariate analysis of variance (MANOVA) was conducted using managers' ratings of budgets and schedules, and technical quality at Time 1 as the dependent variable. This was followed by a second MANOVA in which managers' ratings from Time 2 were used as the dependent variable. MANOVA simultaneously tests the effects of the independent variables on the set of dependent variables while controlling for the type 1 error rate and incorporating the correlations among the independent variables in the test statistic. Following this, path analysis of the model's structural equations was conducted with ordinary least squares estimation of the model's parameters, as the causal ordering of the hypotheses was of interest. Path analysis (James et al., 1984b) was used as it provides the opportunity to test a model with intervening or mediating variables. I did not use LISREL, as variable scores used in this study were aggregated from the individual level to the group level and LISREL cannot deal with two different levels of analysis in one simultaneous step. In addition, the sample size to parameter estimate was less than 5, and the results of LISREL analysis under this condition are unreliable (Breckler, 1990; Bentler, 1985).

For the path analyses, each endogenous-dependent variable was tested with all variables causally proceeding it entered into the regression equation. More specifically, five separate multiple regressions were conducted to determine the direct effects on each of the dependent variables (i.e., Time 1 and Time 2 managers' ratings of budgets and schedules, Time 1 and Time 2 managers' ratings of technical quality, and Time 1 team members' ratings of performance). In each multiple regression, one of the dependent variables was regressed on team size, the exogenous variables (i.e., member time dedication, support and recognition, leader influence, and functional diversity), and external communications and cohesiveness. Then, members' social identification was entered and the incremental change in R^2 was computed. To explore the indirect effects of the independent variables on the dependent variables, three additional multiple regressions were conducted. In the first regression equation, cohesiveness was regressed on team size, the four exogenous variables, and the remaining two process variables. In the second regression equation, external communication was regressed on team size and the four exogenous variables. In the third regression equation, team social identification was regressed on team size and the four exogenous variables.

All three control variables were not tested in the regression equations simultaneously because the small sample size resulted in a loss of statistical power. Rather, the analyses outlined above were repeated two more times substituting technology newness and team tenure for team size.

5. Results

The descriptive statistics and intercorrelations among the study variables are shown in Table 1. Leader influence and social identification significantly correlated with all four of the performance criteria. In addition, support and recognition significantly correlated with three of the four performance criteria. While cohesiveness was not significantly correlated with the managers' ratings of performance in Time 1 or Time 2, it was significantly correlated with the team members' rating of team performance. In contrast, while external communication was not significantly correlated with the team members' ratings of team performance, it was significantly correlated with the managers' ratings in Time 1 and Time 2. Consistent with prior research, the managers' ratings and the team members' ratings were only moderately correlated with each other. Consistent with the hypotheses, (i) leader influence, support and recognition, time dedication, and functional diversity were each significantly correlated in the expected direction with social identification, (ii) functional diversity was significantly correlated with external communications, and (iii) social identification was significantly correlated with cohesiveness. Contrary to hypothesis, external communication was not significantly correlated with cohesiveness.

The scatter diagrams for team tenure, team size, and cohesiveness with each of the four performance criteria were examined to check for any curvilinear relationships. None were in evidence. Further, I added a quadratic term for each of these variables to the respective regression equation and tested for the significance of the incremental change in R^2 (Cohen and Cohen, 1983). Nonsignificant F-tests resulted for all four of the performance criteria suggesting that team tenure, team size, and cohesiveness were linearly related to each of the performance criteria.

The MANOVA result for the Time 1 managers' ratings was significant ($F = 5.058$, $p < 0.001$), as was the result for the Time 2 managers' ratings ($F = 6.322$, $p < 0.001$). Consequently, I proceeded with the path analysis. Table 2 reports the results of regressing each of the four performance criteria on the independent and intervening variables. As shown, the standardized regression coefficient associated with members' social identification was significant in all four regression equations. Further, when social identification was entered into each of the equations after controlling for all the other variables, the incremental change in R^2 was significant, indicating that social identification explained unique variance in each of the performance criteria. In addition, external communications was positively related to both the Time 1 and Time 2 managers' ratings of technical quality and budgets and schedules, but was not significantly rated to the members' ratings of performance. The pattern of findings was reversed for cohesiveness. Cohesiveness was positively related to the members' ratings, but was not significantly related to the Time 1 or Time 2 managers' ratings of quality or budgets and schedules.

Table 3 reports the beta weights resulting from the regression of the three intervening variables on each of the variables causally proceeding them in the theoretical model. This information was used to find the indirect effects of each independent variable on the dependent variables. Fig. 2 shows the empirical model predicting Time 2 budgets and schedules, and Fig. 3 shows the empirical model predicting Time 2 technical quality. Contrary to hypothesis 1, the paths from cohesiveness to technical quality, and

Table 1
Descriptive statistics and pearson product–moment correlations[a]

	Mean	S.D.	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1. Team size	5.02	1.78														
2. Team tenure	10.20	3.61	0.07													
3. Technology newness	3.54	0.97	0.12	0.19												
4. Leader influence	3.85	0.62	0.13	0.19	0.22											
5. Time dedication	0.39	0.11	0.09	0.18	0.22	0.10										
6. Support and recognition	3.75	0.89	0.19	0.18	0.27*	0.01	−0.03									
7. Functional diversity	0.52	0.36	0.27*	0.15	0.19	−0.04	−0.22	0.31*								
8. Social identification	3.08	0.84	0.06	0.23	0.24	0.37*	0.35*	0.30**	−0.31*							
9. Cohesiveness	3.23	0.87	0.13	0.29*	0.18	0.16	0.24	0.37*	−0.28*	0.35*						
10. External communications	3.19	0.89	0.26	0.16	0.28*	0.19	−0.11	−0.04	0.32*	−0.06	0.19					
11. T1 Budgets and schedules	3.46	0.98	0.06	0.18	−0.01	0.48**	0.21	0.46**	−0.23	0.42**	0.13	0.38*				
12. T1 Technical quality	3.12	0.79	0.13	0.16	0.03	0.37*	0.28	0.44**	−0.28	0.48**	0.21	0.41**	0.76***			
13. Team-rated performance	3.87	0.67	0.03	0.21	−0.06	0.36*	0.51**	0.48**	−0.26	0.57**	0.37*	0.14	0.53**	0.41**		
14. T2 Budgets and schedules	3.35	0.85	0.11	0.22	−0.08	0.38*	0.21	0.48**	−0.15	0.39**	0.07	0.39**	0.68***	0.43**	0.35*	
15. T2 Technical quality	3.00	0.71	0.09	0.18	0.09	0.32*	0.27	0.42**	−0.14	0.45**	0.09	0.22	0.41*	0.61**	0.42**	0.72***

[a] $n = 42$.
* $p < 0.05$.
** $p < 0.01$.
*** $p < 0.001$.

Table 2
Results of regression analyses[a]

Independent variables	Dependent variables[b]				
	Manager ratings technical quality (Time 1)	Manager ratings budgets and schedules (Time 1)	Team rating of performance	Manager ratings technical quality (Time 2)	Manager ratings budgets and schedules (Time 2)
1. Team size	0.08	0.07	0.08	0.06	0.03
2. Leader influence	0.19	0.48**	0.18	0.16	0.18
3. Member time dedication	0.25	0.21	0.31*	0.17	0.07
4. Support and recognition	0.26	0.46**	0.15	0.35*	0.31*
5. Functional diversity	−0.28	−0.23	−0.07	−0.26	−0.15
6. External communication	0.35*	0.38*	0.19	0.35*	0.38*
7. Cohesiveness	0.19	0.11	0.34*	0.06	0.06
Adjusted R^2	0.33	0.47	0.38	0.39	0.50
F-Value	3.01*	7.65***	4.63*	4.05*	8.06***
8. Team social identification	0.39*	0.42*	0.48**	0.39**	0.33*
Adjusted R^2	0.38	0.53	0.45	0.44	0.55
F-Value	4.83**	7.98***	7.18***	5.68**	5.58***
Change in R^2	0.06	0.06	0.07	0.05	0.05
F-Value of change	4.33*	4.53*	4.68*	4.05*	4.32*

[a] $n = 42$.
[b] Numbers in columns are beta weights.
* $p < 0.05$.
** $p < 0.01$.
*** $p < 0.001$.

from cohesiveness to budgets and schedules (as rated by managers) were not significant. Hypothesis 2 posited a positive relationship between external communications and the performance criteria. The path analysis supported the hypothesis. The path coefficient from external communications to budgets and schedules was significant ($p = 0.38$, $p < 0.05$), as was the path from external communications to technical quality ($p = 0.35$, $p < 0.05$).

Hypothesis 3 expected a negative relationship between cohesiveness and external communications, and hypothesis 4 expected a positive relationship between cohesiveness and social identification. While cohesiveness was negatively related to external communications, it did not quite reach significance. Thus, hypothesis 3 was not supported. However, cohesiveness was positively related to social identification ($p = 0.38$, $p < 0.01$), consistent with hypothesis 4.

Hypotheses 5, 6, and 7 posited that leader influence, top management support and recognition, and members' time dedication would each be positively related to social identification. The path coefficients' were positive and significant between social identification and support and recognition ($p = 0.39^{**}$, $p < 0.01$), leader influence

Table 3
Results of regression analyses[a]

	Dependent variables[b]		
	Team social identification	External Communication	Cohesiveness
1. Team size	0.13	0.07	−0.16
2. Leader influence	0.36**	0.20	−0.07
3. Member time dedication	0.32*	0.03	0.12
4. Support and recognition	0.39**	−0.12	0.21
5. Functional diversity	−0.29*	0.37*	−0.09
6. Social identification	–	–	0.38**
7. External communications	–	–	−0.26
Adjusted R^2	0.35	0.08	0.29
F-Value	6.59***	2.89	3.78**

[a] $n = 42$.
[b] Numbers in columns are beta weights.
* $p < 0.05$.
** $p < 0.01$.
*** $p < 0.001$.

($p = 0.36^*$, $p < 0.05$), and members' time dedication ($p = 0.32$, $p < 0.05$), indicating support for hypotheses 5, 6, and 7. Further, the path from functional diversity to social identification was negative and significant supporting hypothesis 8 ($p = 0.29$, $p < 0.05$).

Hypothesis 9 anticipated that functional diversity would be positively related to external communications. The positive significant path coefficient from functional

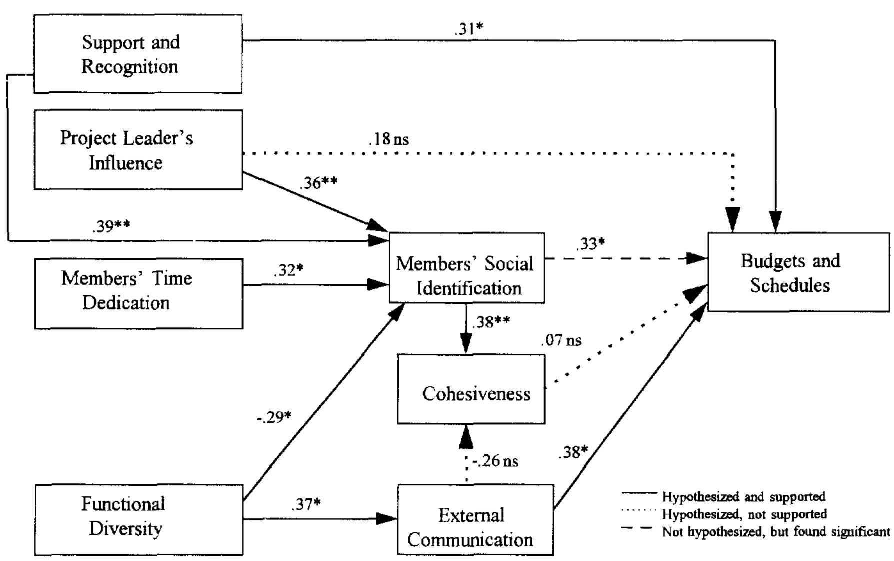

Fig. 2. Path coefficients predicting managers' time 2 ratings of budgets and schedules.

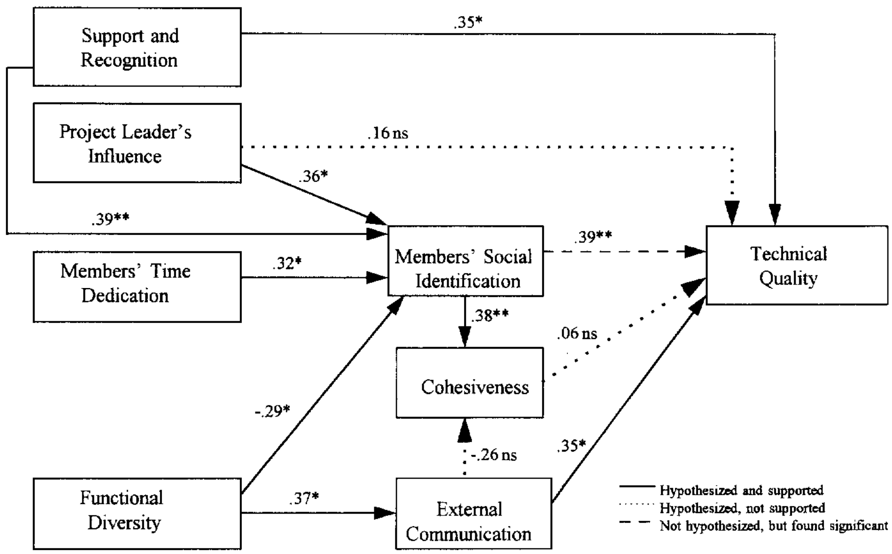

Fig. 3. Path coefficients predicting managers' time 2 ratings of technical quality.

diversity to external communication ($p = 0.37$, $p < 0.05$) offers support for hypothesis 9. Hypothesis 10 was supported by the positive significant path from top management support and recognition to both budgets and schedules ($p = 0.31$, $p < 0.05$) and technical quality ($p = 0.35$, $p < 0.05$). Contrary to hypothesis 11, the path between leader influence and each of the performance criteria was nonsignificant.

To test the overall fit of the model to the data, I calculated a Q statistic comparing the product of all squared residual paths subtracted from one in a fully recursive model to the product of all squared residuals subtracted from one in the over-identified model (Pedhazur, 1982). The Q statistic for the model tested here was 0.91, suggesting a good fit to the data. I also computed the direct and indirect effects of each variable on the endogenous variables in the model. If the paths were properly specified in the model, the analysis should reproduce the correlation matrix with small errors of magnitude (James et al., 1984b). As shown in the last column of Table 4 (column G), the error terms were quite small in magnitude for all paths in the model.

One linkage that was not hypothesized was found empirically significant in both empirical models shown in Figs. 2 and 3. This was the path from members' social identification to budgets and schedules ($p = 0.33$, $p < 0.05$) and technical quality ($p = 0.39$, $p < 0.01$). While I had anticipated that members' social identification would be entirely mediated by cohesiveness, it was directly related to budgets and schedules and technical quality regardless of its relationship with cohesiveness.

To conserve space, the path coefficients for the empirical models are only illustrated here for the managers' ratings in Time 2. However, the pattern of results is very similar to that found using the Time 1 managers' ratings. For both budgets and schedules, team social identification was directly related to the performance criteria, as was external communications. In contrast to the Time 2 rating, leader influence had a direct effect on

Table 4
Decomposition of association between variables for time 2 managers' ratings

Bivariate relationships	Causal effects						
			Indirect effects				
	Total covariance A	Direct effects B	Via social identification C	Via external communications D	Via cohesiveness E	Total effects F = B + C + D + E	Spurious G = A − F
Leader influence with budgets	0.387	0.181	0.123	0.067	−0.015	0.356	0.031
Support and recognition with budgets	0.485	0.356	0.133	−0.043	0.005	0.451	0.034
Time dedication with budgets	0.212	0.073	0.116	0.015	−0.016	0.188	0.024
Functional diversity with budgets	−0.143	−0.155	−0.093	0.113	0.041	0.148	−0.005
Social identification with budgets	0.399	0.334	–	–	0.021	0.345	0.045
External communications with budgets	0.390	0.384	–	–	−0.011	0.379	0.011
Cohesiveness with budgets	0.072	0.041	–	–	–	0.043	0.031
Leader influence with technical quality	0.324	0.164	0.143	0.053	−0.012	0.348	−0.024
Support and recognition with technical quality	0.426	0.312	0.156	−0.034	0.009	0.443	−0.017
Time dedication with technical quality	0.271	0.178	0.127	0.015	−0.012	0.308	−0.037
Functional diversity with technical quality	−0.288	−0.263	−0.117	0.145	−0.053	−0.286	−0.002
Social identification with technical quality	0.459	0.397	–	–	0.016	0.413	0.046
External communication with technical quality	0.363	0.354	–	–	−0.021	0.334	0.029
Cohesiveness with technical quality	0.092	0.063	–	–	–	0.063	0.029

Time 1 budgets and schedules ($p = 0.48$, $p < 0.01$), as well as the indirect effect through social identification. Further, the direct effect from support and recognition to social identification found with the Time 2 managers' ratings was not found with the Time 1 data.

The pattern of findings for the team members' rating was quite different than that found for the managers' ratings in both Time 1 and Time 2. The path from cohesiveness to member-rated performance was significant ($p = 0.34$, $p < 0.05$). While social identification had a positive direct effect on member rated performance ($p = 0.48$, $p < 0.01$), it also had an indirect effect through cohesiveness. The paths to member-rated performance from external communications, support and recognition, and from leader influence, were not significant. Finally, member time dedication had a direct effect on member-rated performance ($p = 0.31$, $p < 0.05$), as well as the indirect effect through social identification.

In each of the analyses reported above, the tests of the hypotheses were conducted after partialling the effects of team size from the regression equation. The other control variables—technology newness and team tenure—were not included in the model simultaneously with team size to maintain adequate power to test the model given the small sample size ($n = 42$) (Cohen and Cohen, 1983). As seen in Table 1, neither of these variables showed a strong pattern of correlation with the other study variables. When the analyses were repeated using each of the alternative control variables in place of size, the results were entirely consistent with those reported here.

Two additional exploratory analyses were conducted. As presented above, I found an unanticipated positive direct effect of social identification on each of the performance variables. Given that social identification often results in a partitioning of the social world into an 'in-group' and 'out-group', it seemed possible that highly socially identified teams might have poor intergroup relations that could negatively affect their performance. To test this possibility, I repeated the analyses adding a quadratic term in each of the regression equations and testing for the significance of the incremental change in R^2 (Cohen and Cohen, 1983). Nonsignificant F-tests resulted for all four of the performance criteria, indicating that member social identification was linearly related to each of the performance criteria studied here.

Finally, the sample consisted of both product and process development teams. I had no reason to propose that any of the relationships specified in the model would be moderated by the type of project, but I tested for interaction effects in the interest of completeness. I repeated the analyses introducing a dummy-coded variable representing project type into the regression equations. Again, to maintain statistical power, I substituted the dummy-coded variable for team size. The test of the overall regression equation remained significant ($F = 6.15$, $p < 0.001$), and the regression coefficient associated with the dummy variable was not significant. I could not simultaneously assess the interaction of each of the independent and intervening variables with the dummy variable due to the loss of statistical power, so I examined each independently through hierarchical regression. I first entered the independent variable and the dummy variable into the regression equation followed by the interaction term (composed of the product of the independent and dummy variable). I then computed the incremental change in R^2 generated by the inclusion of the interaction term. The results of all tests were not significant, indicating that none of the hypothesized relationships differed between the two types of development projects.

6. Discussion

Empirical research on group effectiveness has not often investigated group members' social identification—the extent to which members are psychologically identified with a group. In the current study, I suggested that characteristics of project teams might make identity issues of particular concern. These include their functional diversity, limited temporal scope, and the nonroutine nature of their tasks. Drawing from social identity theory and group effectiveness research, this study presented a model of team effectiveness in which members' social identification, cohesiveness, and the amount of external communications were posited to jointly affect team performance. This model was tested in a sample of 42 development teams, and the results suggest that members' social identification is, in fact, a critical influence on team performance, at least in the R&D setting studied here.

Team members clearly differentiated between social identification and cohesiveness. While the two variables were positively related to each other, the correlation between them was modest in size, and they were differentially related to the other variables in this study. Further, the path between social identification and cohesiveness was positive and significant. While the current study is cross-sectional in design and cannot address causation, the results of the path analysis are consistent with the theoretical propositions of social identity theory. That is, cohesiveness appeared to result from social identification rather than the other way around.

At the midpoint of the development projects, team members' social identification was positively related to member-rated performance. Social identification had both direct effects on member-rated performance, as well as indirect effects through cohesiveness. Consistent with prior studies with sales groups (Gladstein, 1984) and development teams (Ancona and Caldwell, 1992b), members tended to rate team performance higher under conditions of internal team cohesiveness. In contrast, the relationship between cohesiveness and managers' ratings of performance at both project midpoint and project completion was not significant. The nonsignificant path between cohesiveness and the managers' performance ratings is consistent with some prior studies in development teams (e.g., Ancona and Caldwell, 1992a,b) while being at odds with other studies (e.g., Keller, 1986). Goodman et al. (1987), reviewing the literature on group effectiveness, argued that there is no evidence for a link between cohesiveness and performance. If the criterion of effectiveness is restricted to the managers' ratings of performance, the results here support this view.

While I had anticipated that the link between social identification and performance would be entirely mediated by cohesiveness, the results showed a positive direct path between social identification and manager-rated team performance regardless of the level of cohesiveness. Social identification did result in higher cohesiveness, yet this was not the mechanism through which it influenced manager-rated performance. Why simply identifying with the team yielded higher performance results remains an interesting question.

One explanation may lie in the motivational aspects of social identification. Turner (1982) speculated that even as group members strive to positively differentiate their group from other groups, they also seek to positively differentiate themselves from other group members by striving to enact the positive attributes to a greater degree than do other group members. In other words, group members compete with each other 'to be the first among equals' in enactment of prototypical group characteristics. McGrath (1996) noted that projects consist of multiple tasks, subsets of activities that cumulatively contribute to project completion. Within a project, an individual group member undertakes tasks or activities that are frequently only loosely coupled to the activities of other group members. Loose coupling means that members have some degrees of freedom in their actions, as not all activity needs to be temporally coordinated with other members. Perhaps social identification increases group performance through heightened effort of individual members on independent tasks or activities. Unfortunately, the group literature has tended to concentrate on group tasks rather than group projects per se (McGrath, 1996), so we know relatively little as yet about how teams combine separate activities of members over time.

Another possibility for the positive direct relationship between members' social identification and performance criteria is that it is a statistical artifact resulting from a missing intervening variable. Research on social dilemmas has shown that socially identified members engage in behaviors that benefit the collective rather than their own self-interest (Brewer and Kramer, 1986; Kramer and Brewer, 1984). Rather than operating through cohesiveness or improved interpersonal relations among the team members, social identification may positively influence internal task processes such as the setting of common goals, and the determination of task priorities. Unfortunately, the data were not available to test this.

Consistent with prior research, external communications was positively related to manager-rated performance. Functional diversity contributed to increasing external communications, but also had a negative indirect effect on manager-rated performance through members' social identification. Ancona and Caldwell (1992b) speculated that their findings of a direct negative effect of functional diversity on development team performance may have been the result of a missing intervening variable in their theoretical model. The findings here suggest that social identification intervenes between functional diversity and manager-rated performance.

Top management support and recognition had a direct positive effect on manager-rated performance, as well as indirect positive effects through social identification. The effect of leader influence on manager-rated performance at project completion was entirely indirect through social identification. Further, I found that members' social identification increased as the proportion of team members' time allocated to the project increased. The effects of this variable on manager-rated team performance were entirely mediated by the members' social identification.

Prior studies have shown top management support and recognition and project leader influence related to development team performance (e.g., Clark and Fujimoto, 1991; Katz and Allen, 1985), but they have not investigated the mechanism through which these variables operate. The results reported here suggest that top management support and recognition is of both instrumental and symbolic value to project teams. Further, in the case of leader influence and team members' time dedication, the lack of direct effects indicates that these variables operate on performance primarily by influencing team members' social identification. I speculate that each of these enhances the attractiveness of team membership by serving to distinguish the project as an important or prestigious enterprise within the organization.

The overall model explained significant variance (i.e., from 38 to 55%) in the performance of the development teams studied here. The inclusion of social identification in the model added significant explanatory power over and above that provided by variables that had been investigated previously in the literature.

7. Implications, limitations, and directions for future research

7.1. Theoretical and practical implications

The theoretical value of this study is that it establishes the relevance of social identity theory in the context of small project groups. In the small face-to-face teams studied here, the basis of group identification did not appear to be personalized bonds of attachment among team members but impersonal bonds derived from the common identity of 'project team member'. While researchers have discussed the effects of categorization on identity formation processes previously (e.g., Kramer, 1991), the focus of much of this work has been on intergroup conflict rather than on intragroup effectiveness. Theory on group dynamics has stressed boundary management as important to identity formation (e.g., Alderfer, 1987; Friedlander, 1987), but group effectiveness research has seldom empirically examined the determinants and outcomes of members' social identification on group effectiveness in organizational settings. Most significantly, the study tends to focus attention on the social identification effects of symbolic aspects of team functioning, which have not been given much attention in the group effectiveness literature. At the organizational level, we have long realized that management action is as much symbolic in its effects as it is substantive (Pfeffer, 1981), but we have been relatively slow to investigate these symbolic effects at the subgroup level.

From the perspective of managers, the current study findings suggest that simply calling a collection of people 'a project team' is inviting poor performance unless steps are taken to give the categorization a meaning in the eyes of the participants. Project managers and top management need to understand the potential self-enhancement effects of categorizations on team members, and attend carefully to aspects of team structure and context for projects where high member social identification is desired. This appears to be particularly important when teams are constituted of members from different functional areas. Others have noted the negative effects of cross-functional diversity on a teams' ability to develop a shared perspective or a shared view of reality (Dougherty, 1992), and this is likely to be exacerbated when cross-functional members fail to psychologically integrate project team membership in the construal of their individual identities. The findings here show that powerful project leaders, top management support and recognition, and allocating members full-time to the team are useful mechanisms to affect the members' social identification with the team.

7.2. Limitations

The findings of the current study should be interpreted with caution. All of the teams studied here were development teams. Despite the fact that the hypotheses were theoretically derived from social identity theory, generality of the results to other types of small face-to-face groups remains in question. Further, common method variance is a potential problem in the findings reported for member-rated performance. Also, the relatively small sample size limited my ability to include other sources of variation in development team effectiveness that have been considered in the literature. Even with the small subset of control variables considered here, I could not test all effects simultaneously.

Although I took care to assure that aggregation of individual responses to the group level was appropriate, aggregation biases and statistical artifacts could still influence the results. In addition, the study design uses cross-sectional data to test the hypotheses limiting the ability to infer causation. Finally, a note of caution is in order relative to the path analysis. The results shown here test only one theoretical path model. Any number of models may have fit the data (Pedhazur, 1982).

7.3. Directions for future research

This study underscores the importance of team identification to ad hoc project teams, at least in the product and process development context studied here. While the results are interesting and useful, additional research is needed to strengthen confidence in the theoretical model presented here. There is also significant opportunity to identify additional influences on the social identification of team members. Longitudinal studies would be very useful to determine the casual relationships in the model. Finally, the current study was strictly concerned with the benefits of intrateam social identification on team effectiveness. However, top managers must consider the projects within the larger organizational context, and be concerned about the effects intrateam social identification might have on intergroup relations (cf., Kramer, 1991). There is considerable opportunity to investigate these intra- and intergroup processes simultaneously.

In summary, the findings reported here provide a more comprehensive understanding of project team effectiveness and suggest that the study of organizational teams should consider the effects of the members' social identification on team performance. I hope future researchers will continue to explore additional aspects of team context and structure that might influence social identification to provide more useful prescriptions for practising managers as they seek to develop high-performing project groups.

Appendix A. Rotated factor matrix of social identification and cohesiveness items (2-Factor Constraint)

Question no.	Social identification	Cohesiveness
6	0.78476	0.10791
2	0.76582	0.17600
1	0.72115	0.09591
3	0.72021	0.11913
5	0.67543	0.19235
7	0.62944	0.10437
4	0.59453	0.19818
24	0.06458	0.74311
22	0.11983	0.70071
21	0.07629	0.67641
23	0.23688	0.53169

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